

CodeGalaxy

An Agentic AI Based Interactive Programming Learning Platform

B.Tech Major Project Synopsis — 2026–2027

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Abstract

CodeGalaxy is an Agentic AI-powered interactive programming learning platform designed to help beginners understand programming concepts through visualization and intelligent guidance. Traditional learning methods often focus on syntax and output but fail to explain the internal execution of programs. CodeGalaxy allows users to write code in languages like Java, Python, and JavaScript, while AI agents analyze, explain, and visualize the complete program execution process.

The platform uses multiple AI agents, including a Code Analysis Agent for detecting errors and suggesting improvements, a Visualization Agent for showing real-time execution steps, and a Learning Assistant Agent for providing personalized guidance. The Assessment Agent evaluates user skills and generates suitable coding challenges based on performance.

By combining Agentic AI, real-time visualization, and gamification features such as levels, rewards, badges, and progress tracking, CodeGalaxy creates an engaging learning environment. It helps students build strong programming concepts, improve problem-solving skills, and learn coding through AI-driven assistance instead of only memorizing programs.

Problem Statement

Many beginners face difficulties in understanding programming concepts because traditional learning methods mainly focus on syntax and output rather than explaining the internal working of programs. Complex topics such as loops, recursion, algorithms, data structures, and memory management are difficult to learn without proper visualization of program execution. Existing coding platforms provide limited support by only showing errors or final outputs, making it challenging for students to identify logical mistakes and understand the flow of code. Additionally, traditional learning approaches do not provide personalized guidance based on individual performance, which reduces learning effectiveness and

motivation. Therefore, there is a need for an Agentic AI-based programming platform that can analyze user code, explain execution steps, visualize programming concepts in real time, provide intelligent feedback, generate personalized challenges, and create an engaging learning environment through gamification. CodeGalaxy aims to solve these problems by combining AI-driven assistance, visualization, and interactive learning to help beginners develop strong programming skills and conceptual understanding.

Objectives

The main objective of CodeGalaxy is to develop an Agentic AI-powered interactive learning platform that helps beginners understand programming concepts through intelligent guidance, real-time visualization, and personalized learning. The application aims to analyze user-written code, identify errors, explain program execution steps, and visually represent concepts such as loops, data structures, algorithms, and memory operations to improve conceptual understanding. It also focuses on providing an AI-based programming mentor that assists learners by answering queries, suggesting improvements, and adapting learning content according to user performance. By integrating gamification features such as coding challenges, rewards, levels, and progress tracking, CodeGalaxy aims to make programming learning more engaging, effective, and interactive while improving problem-solving skills and reducing the dependency on memorizing code.

System Architecture

Module 1: User Profile and Learning Management Module:

The User Profile Module manages user details and learning activities such as username, email, profile information, selected programming languages, completed challenges, badges, scores, levels, and coding history. The Agentic AI Learning Agent continuously analyzes user activities and learning patterns to understand individual strengths and weaknesses. Based on user performance, the AI agent provides personalized learning recommendations, suggests suitable coding challenges, and adapts the learning path according to the user's skill level.

Module 2: AI-Powered Code Editor Module:

The Code Editor Module allows users to write, edit, and practice programming languages such as Java, Python, and JavaScript. The module provides an interactive coding environment where users can execute programs and receive intelligent assistance. The Code Assistance Agent analyzes the written code in real time, identifies syntax errors, logical mistakes, and inefficient coding practices. It provides explanations, improvement suggestions, and coding guidance like an AI programming mentor.

Module 3: Intelligent Execution Engine Module:

The Execution Engine Module is responsible for processing and executing user-written programs step by step. The Agentic AI Execution Agent monitors the complete execution process, analyzes program behavior, tracks variable changes, memory usage, loops, and conditions. It explains how each instruction is executed and provides meaningful insights instead of only displaying the final output.

Module 4: AI-Based Visualization Module:

The Visualization Module converts program execution into interactive graphical representations. It displays concepts such as variables, arrays, linked lists, stacks, queues, recursion, sorting algorithms, and searching algorithms. The Visualization Agent understands the execution flow and automatically generates visual explanations to help learners understand complex programming concepts easily.

Module 5: AI Learning Assistant Module:

The Learning Assistant Module works as a virtual programming tutor for users. The AI Mentor Agent interacts with learners, answers programming-related questions, explains difficult concepts, provides examples, and guides users during problem-solving.

Module 5: Gamification and Challenge Module:

The Gamification Module improves user engagement by providing levels, rewards, badges, experience points, coding challenges, and achievements. The Challenge Generation Agent analyzes user performance and automatically creates personalized coding tasks with suitable difficulty levels. The AI agent increases challenge complexity as the user's skills improve.

Module 5: Database and Knowledge Management Module:

The Database Module securely stores user profiles, coding history, progress data, completed challenges, scores, AI-generated recommendations, and learning reports. The AI Knowledge Management Agent uses stored learning data to improve recommendations and maintain personalized learning experiences for users.

Module 5: Performance Analytics Module:

The Analytics Module tracks user coding activity, learning progress, problem-solving ability, and performance reports. The Analytics Agent analyzes user behavior, identifies weak areas, predicts learning difficulties, and provides improvement suggestions to enhance programming skills.

Module 5: Admin Management Module:

The Admin Module manages users, coding challenges, learning content, badges, system settings, and platform activities. The Admin AI Agent helps administrators by analyzing platform usage, monitoring user engagement, generating reports, and suggesting improvements for better system performance.

Technology Stack

Component	Technology
Frontend Development	React.js, HTML5, CSS3, JavaScript
Backend Development	Node.js, Express.js, Python
Agentic AI Framework	LangChain, OpenAI API, AI Agents
Code Execution Engine	Docker, Sandbox Environment, Compiler APIs
Data Visualization	D3.js, React Flow, Three.js
Database	MongoDB
Authentication & Security	JWT Authentication, bcrypt
AI Learning Assistant	Large Language Model (LLM), NLP
Development Tools	VS Code, PyCharm
Version Control	Git, GitHub
Deployment	AWS / Vercel / Render
Browser Support	Google Chrome, Mozilla Firefox, Microsoft Edge

Expected Outcomes

The expected output of the CodeGalaxy – Agentic AI Based Interactive Programming Learning Platform is an intelligent and interactive environment that helps users learn programming concepts effectively. Users will be able to write code in programming languages such as Java, Python, and JavaScript through an integrated code editor. The system will analyze the code using AI agents, detect syntax and logical errors, provide real-time suggestions, and execute the program step by step.

The platform will generate visual representations of program execution, allowing users to observe variable changes, loop iterations, conditional statements, memory allocation, and data structure operations in real time. An AI Learning Assistant will provide explanations, answer programming-related questions, and offer personalized guidance based on the learner's performance and learning behavior.

The system will also provide coding challenges, rewards, badges, experience points, and progress tracking through a gamified learning environment. AI agents will continuously monitor user performance, identify weak areas, recommend suitable learning materials, and generate personalized coding exercises. Administrators will be able to manage users, learning content, challenges, and system activities through a dedicated admin panel. Overall, the platform is expected to improve programming understanding, enhance problem-solving skills, increase learner engagement, and provide a personalized coding education experience through Agentic AI.

Conclusion

CodeGalaxy – An Agentic AI Based Interactive Programming Learning Platform provides an innovative approach to programming education by combining artificial intelligence, real-time visualization, and gamification. The platform addresses the limitations of traditional

programming learning methods by helping beginners understand the internal execution of programs instead of only focusing on syntax and output. Through AI agents, the system can analyze code, detect errors, provide personalized guidance, visualize execution steps, and recommend suitable learning activities based on user performance.

The integration of Agentic AI makes CodeGalaxy an intelligent learning environment where users receive continuous assistance from AI-based mentors, execution analyzers, and recommendation systems. Features such as interactive visualization, coding challenges, rewards, progress tracking, and adaptive learning improve user engagement and make programming more understandable and effective.

Overall, CodeGalaxy aims to transform programming education into a more interactive, personalized, and practical experience. By reducing the difficulty of complex programming concepts and encouraging problem-solving skills, the platform helps learners build strong programming foundations and develop confidence in coding through AI-driven learning support.